CellTrackAnalysis Instructions

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This is the **second step** of chemotaxis analysis in TumbleWeed. `CellTrackAnalysis.m` reads in the X and Y coordinates of bacteria (the output from Step 1, `SimpleTracking.m`) and provides several chemotaxis metrics including tumble frequency of the experimental group, and as well as individual bacterium's of the group.

- Input: `SimpleTrackingoutput.mat` (Output file from Step 1, `SimpleTracking.m`)
- Outputs
    - `CellTrackAnalysis_results.xlsx` (Excel file with 2 tabs)
        - PerTrack_Metrics: Metrics per bacterium
        - Metadata: Summarized metrics of the group
    - Printout of summary metrics of the experimental group
    - Visualizations of metrics' distributions

In this step, from the tracked X and Y, the linear and angular velocity of each bacterium are calculated. Then, the algorithm finds the local minima of the linear velocity and local maxima of angular velocity and finds the tumble points based on the calculation under user defined threshold (Johnson et al., 2024).

How to use and interpret CellTrackAnalysis

1. Open MATLAB, click "Browse For Folder" icon to navigate to TumbleWeed folder
2. Find and open `CellTrackAnalysis.m`
3. Adjust parameters if needed. The most important parameters are listed in the table below. Recommendations for adjusting the parameters:
    a. Important: Make sure `secondVideo` is correct.
    b. Run the script with **default** value first, and then validate. You can quickly validate the results in Step 3 (`PlotTrajectory.m`) with a setting of `bacteria_idx = 0`. If the values do not make sense, come back to Step 2 and adjust `depth_speed_ratio`.
    c. If the algorithm is still displaying too many tumbles, try increasing `sg_window` with `fps_scale_sg_window=false`. Only odd numbers can be used for `sg_window`.

Parameter Name	Value Range	Data Type	Description
`secondVideo`	> 0	Integer	Duration of the video in second.
`depth_speed_ratio`	0.5 – 0.9999	Float	The ratio of linear speed vs local minima of the speed. The larger value will return less tumbles (more conservative). For more details, refer to Johnson et al. (2024)
`show_bacteria_labels`	true / false	Boolean	Setting this true will display bacteria ID numbers on the scatterplot output.
`fps_scale_sg_window`	true / false	Boolean	Setting this true will auto scale `sg_window` based on the video's FPS value. If set to false, will use the `sg_window` value input.
`sg_window`	Odd number	Integer	Window size in the Savitzky-Golay filter. A larger value will return less tumbles (more conservative). Highly recommend trying the `fps_scale_sq_window = true` then think about adjusting from auto-scaled number afterwards. For more details, Refer to sgolay function description in MATLAB.
`convert_to_um`	true / false	Boolean	Setting this to true will convert pixels to um.
`px_to_um`	> 0	Float	Conversion value: um per pixel
`min_tumble_speed`	> 0	Float	Minimum speed at tumble dip

4. Histograms

a. After running the script, TumbleWeed provides distributions of the metrics such as tumble count, tumble and run duration, tumble angle, linear speed, migration and track duration in histogram.

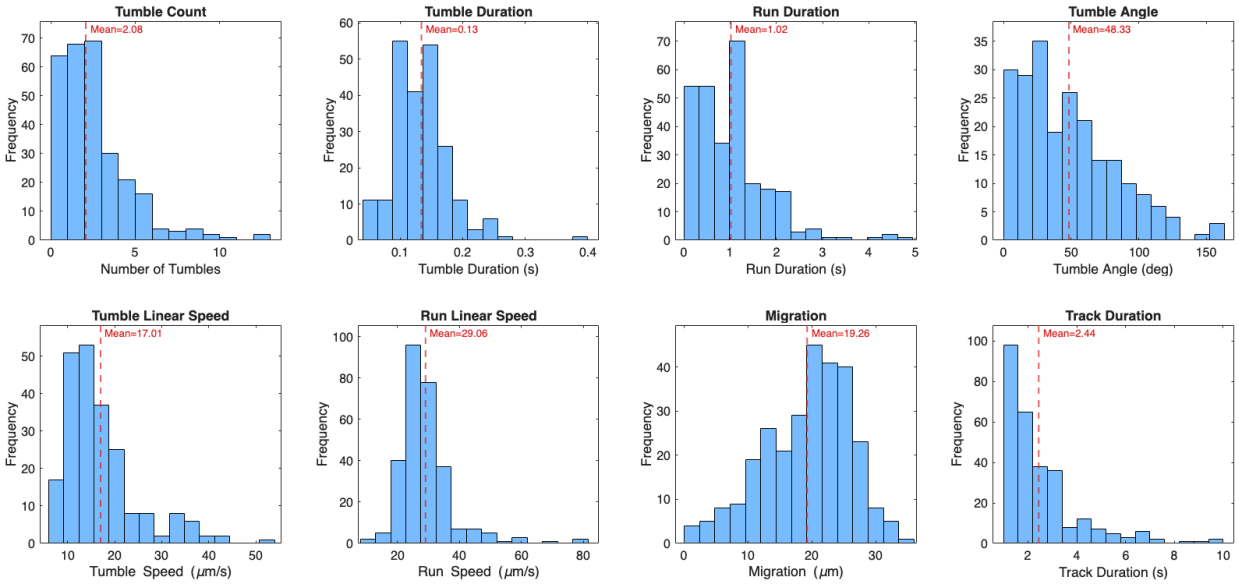
b. The histograms are a good way to validate if the parameters are set correctly. For example, keep an eye on:

i. Likely **High** in chemorepellent (opposite in chemoattractant):

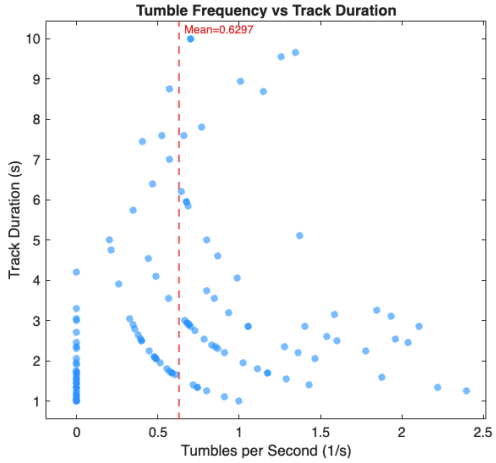
1. Tumble Count, Tumble Duration, Tumble Angle

ii. Likely **Low** in chemorepellent (opposite in chemoattractant):

1. Run Duration, Run Linear Speed, Migration, Track Duration



5. **Scatterplot: Tumble frequency vs track duration**
    a. After running the script, TumbleWeed provides a scatterplot of tumble frequency vs track duration. You can choose whether to show the track IDs on the display by setting `show_bacteria_labels` as true or false.
    b. The scatterplots are great for detecting any outliers, anomalies or looking into a specific track ID.



6. If the parameters are verified already and you're running the script repeatedly for bulk data, setting `show_plots` as false will save the execution time and won't show any displays.
7. **Summary Metrics**
    a. TumbleWeed also prints out the summary metrics of the experimental group.

```
========================================================================
          Population Summary  (N = 131 bacteria analysed)
========================================================================
Metric                                          Mean       Median
------------------------------------------------------------------------
Number of tumbles                               2.11         1.00
Tumbles per second (1/s)                        0.63         0.57
Tumble duration (s)                             0.13         0.12
Run duration (s)                                1.27         1.05
Tumble angle (deg)                             44.86        38.62
Tumble linear speed (µm/s)                     12.88        12.39
Run linear speed (µm/s)                        22.98        22.25
Track duration (s)                              3.02         2.35
========================================================================
```

8. **Excel Output:** `CellTrackAnalysis_results.xlsx`
    a. PerTrack_Metrics: Metrics per bacterium

TrackID	TumbleCount	Tumbles_per_s	MeanTumbleDuration_s	MeanRunDuration_s	MeanTumbleAngle_deg	MeanTumbleSpeed_um_per_s	MeanRunSpeed_um_per_s	TotalLinearSpeed_um_per_s	Migration_um	TrackDuration_s	FirstSecondDisplacement_um	FirstSecondSpeed_um_per_s
1	3	0.3	0.083333333	1.175	36.05333333	11.37473579	28.44884008	28.01767583	10.98255957	10	22.87012075	27.02818844
2	0	0		2			29.52936055	29.52936055	35.37717579	2	26.17542861	28.60175368
3	0	0		1.15			28.01748806	28.01748806	26.6823599	1.15	23.67627259	25.4872702
4	0	0		2.4			28.60468128	28.60468128	45.86877648	2.4	25.39954299	27.7429905
5	0	0		1.2			29.14159625	29.14159625	31.92851718	1.2	27.77683666	28.41635404
6	0	0		5.4			32.05454448	32.05454448	65.12352993	5.4	27.40550948	30.14282647
7	1	0.10416667	0.15	9.45	0.22	16.54542282	29.56690138	29.36129909	244.0123617	9.6	27.93875155	28.37419042

    b. Metadata: Summarized metrics of the group

Row	NumberOfTumbles	TumblesPerSecond_1_per_s	TumbleDuration_s	RunDuration_s	TumbleAngle_deg	TumbleLinearSpeed_um_per_s	RunLinearSpeed_um_per_s	TotalLinearSpeed_um_per_s	Migration_um	TrackDuration_s
Mean	0.508379888	0.207625455	0.112719298	1.947988827	32.48638889	16.79993441	30.34956401	30.0602756	46.72364967	2.377094972
Median	0	0	0.1	1.35	21.69	15.75920422	30.78637214	30.58266479	37.85335095	1.65
StdDev	0.932368707	0.396114534	0.04476862	1.700528051	36.67115201	7.001205836	5.354065471	5.496160503	33.93025115	2.117617499
Variance	0.869311405	0.156906724	0.002004229	2.891795652	1344.77339	49.01688316	28.66601706	30.20778027	1151.261943	4.484303873
SEM	0.069688509	0.02960699	0.005929749	0.127103434	4.85721274	0.927332367	0.400181642	0.410802324	2.536066042	0.158278163

9. Full list of variables

Parameter Name	Value Range	Data Type	Description
secondVideo	> 0	Integer	Duration of the video in second.
convert_to_um	true / false	Boolean	Setting this to true will convert pixels to um.
px_to_um	> 0	Float	Conversion value: um per pixel
fps_scale_sg_window	true / false	Boolean	Setting this true will auto scale sg_window based on the video's FPS value. If set to false, will use the sg_window value input.
sg_window	Odd number	Integer	Window size in the Savitzky-Golay filter. A larger value will return less tumbles (more conservative). Highly recommend trying the fps_scale_sq_window = true then think about adjusting from auto-scaled number afterwards. For more details, Refer to sgolay function description in MATLAB.
sg_poly	2	Integer	Polynomial order of the function (keep at 2)
minTrackLength	sg_window + 2	Integer	Minimum track length to include in the calculation
w_offset	> 0	Integer	Rad/s offset added to the filter for overlay plot
depth_speed_ratio	0.5 – 0.9999	Float	The ratio of linear speed vs local minima of the speed. The larger value will return less tumbles (more conservative). For more details, refer to Johnson et al. (2024)
v_min_threshold	> 0	Float	Tumble frame width threshold Johnson et al. (2024)
use_angular_confirmation	true/ false	Boolean	Requires angular speed at the tumble frame to exceed the track-mean angular speed
w_confirm_factor	> 0	Float	The confirm factor number for angular speed at the tumble frame. Default value is 0.2. Johnson et al. (2024)
min_tumble_speed	> 0	Float	Minimum speed at tumble dip
show_bacteria_labels	true / false	Boolean	Setting this true will display bacteria ID numbers on the scatterplot output.
show_plots	true/false	Boolean	Setting this true will display two visualizations (scatterplot of tumble/s, histograms of main metrics). Set this to false for faster bulk processing.

Reference

- Johnson, S., Freedman, B. & Tang, J. X. Run-and-tumble kinematics of *Enterobacter* Sp. SM3. *Phys. Rev. E* **109**, 064402 (2024).
- Cell Track Analysis Algorithm GitHub: https://github.com/Tang-BioPhys/SM3-Motility/tree/main